COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31s2
DATE PERFORMED:	10/09/2026	**DATE SUBMITTED:**	10/09/2026
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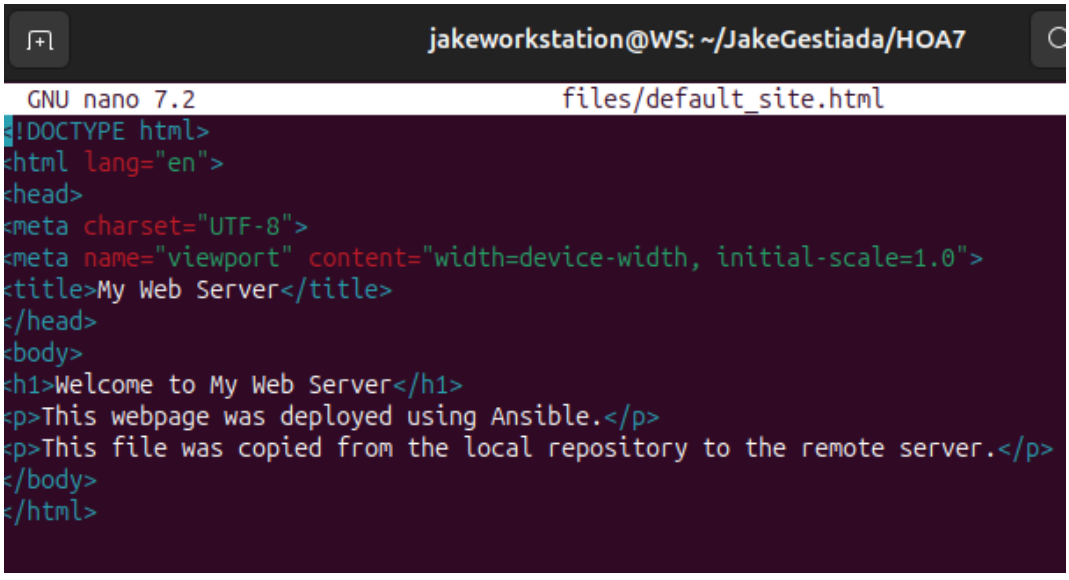
ACTIVITY NO. 7.1 **Managing Files and Creating Roles in Ansible**

1. PROCEDURE OUTPUT

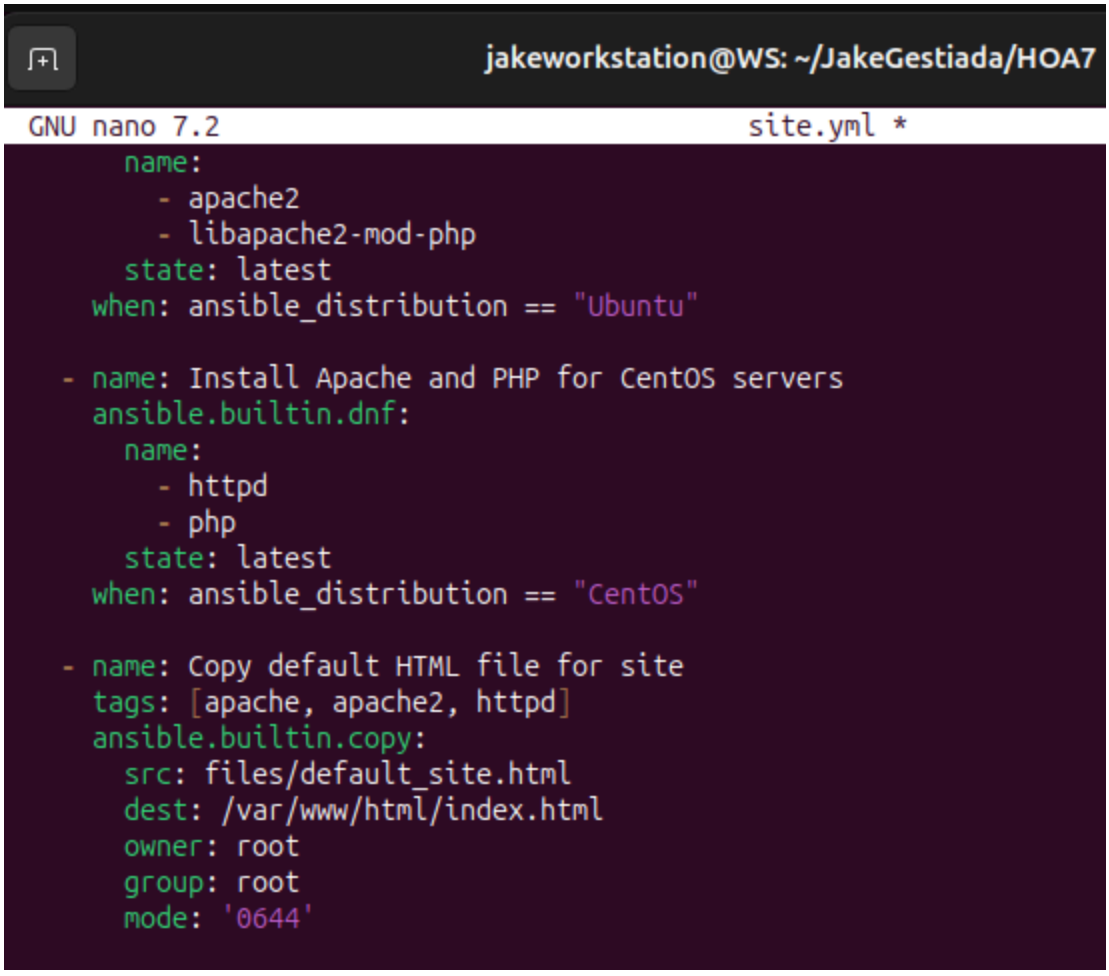
What procedures did you perform?

Task 1: Create a file and copy it to remote servers

1. Using the previous directory was created in your previous laboratory, create a directory, and name it “files.”Create a file inside that directory and name it “default_site.html.” Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit



2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for site:




```
jakeworkstation@WS:~/JakeGestiada/HOA7$ ansible-playbook -i inventory.yaml site.yml --syntax-check

playbook: site.yml
```

3. Run the playbook site.yml. Describe the changes.

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ ansible-playbook -i inventory.yaml site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Update repository index (CentOS)] ****************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [Update package cache (Ubuntu)] *******************************************************
skipping: [centos1]
ok: [ubuntu2]
ok: [ubuntu1]

PLAY [web_servers] *************************************************************************

TASK [Gathering Facts] *********************************************************************
```

```
TASK [Install MariaDB package (Ubuntu)] ****************************************************
skipping: [centos1]
ok: [ubuntu2]

PLAY [file_servers] ************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu1]

TASK [Install Samba package] ***************************************************************
ok: [ubuntu1]

PLAY RECAP *********************************************************************************
centos1                    : ok=8    changed=1    unreachable=0    failed=0    skipped=3    rescued=0
   ignored=0
ubuntu1                    : ok=7    changed=1    unreachable=0    failed=0    skipped=2    rescued=0
   ignored=0
ubuntu2                    : ok=8    changed=1    unreachable=0    failed=0    skipped=3    rescued=0
   ignored=0
```

Observation:

- The plan was put into action and the HTML page name was uploaded the web servers as index.html. The web file also got stored in the correct web folder with a reference so that the servers will be able to load the file as their main homepage.

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS servers. On the CentOS server, go to the browser and type its IP address. Describe the output.

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ ssh jakeserver@192.168.56.105
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

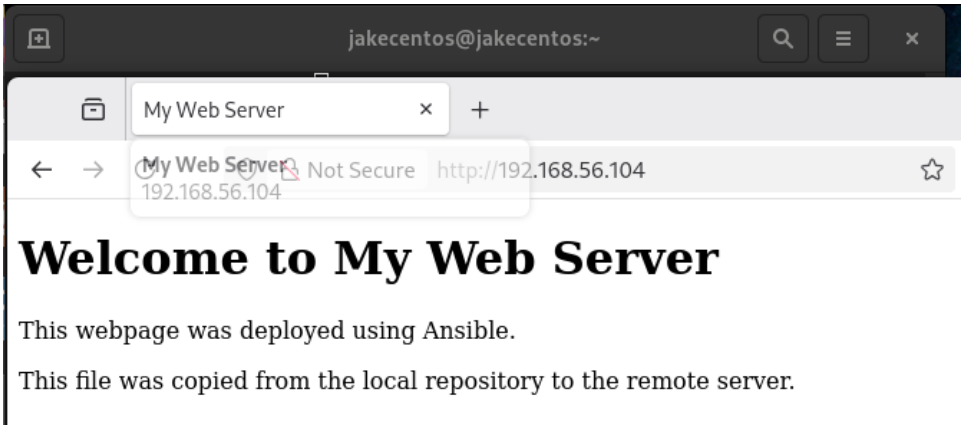
43 updates can be applied immediately.
11 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Sep 10 09:07:29 2026 from 192.168.56.101
jakeserver@server1:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Web Server</title>
</head>
<body>
<h1>Welcome to My Web Server</h1>
<p>This webpage was deployed using Ansible.</p>
<p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
```

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ ssh jakecentos@192.168.56.104
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Sep 10 09:07:24 2026 from 192.168.56.101
[jakecentos@jakecentos ~]$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Web Server</title>
</head>
<body>
<h1>Welcome to My Web Server</h1>
<p>This webpage was deployed using Ansible.</p>
<p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
[jakecentos@jakecentos ~]$
```



Observation:

- I checked the index.html file on the Ubuntu and CentOS servers, and it matched the original HTML file. When I opened the CentOS IP address in the browser, the webpage was displayed correctly.

5. Sync your local repository with GitHub and describe the changes.

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ git commit -m "Task 1"
[main f389542] Task 1
 3 files changed, 128 insertions(+)
 create mode 100644 HOA7/files/default_site.html
 create mode 100644 HOA7/inventory.yaml
 create mode 100644 HOA7/site.yml
jakeworkstation@WS:~/JakeGestiada/HOA7$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 3 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 1.29 KiB | 264.00 KiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:DamienJakeGestiada/CPE212_JakeGestiada.git
   7f89b87..f389542  main -> main
jakeworkstation@WS:~/JakeGestiada/HOA7$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Observation:

- I synced my changes with GitHub, so the new Activity 7 files were uploaded and saved in the repository.

Task 2: Download a file and extract it to a remote server

1. Edit the site.yml. Just before the web_servers play, create a new play:

note: In Ansible, a play is a section of a playbook that defines which remote machines Ansible should manage and what tasks it should perform on them

```
jakeworkstation@WS: ~/JakeGestiada/HOA7
GNU nano 7.2                                    site.yml
    changed_when: false
    when: ansible_distribution == "Ubuntu"


- hosts: workstations
  become: true

  tasks:

    - name: Install unzip
      ansible.builtin.package:
        name: unzip
        state: present

    - name: Install Terraform
      ansible.builtin.unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
        dest: /usr/local/bin
        remote_src: yes
        mode: '0755'
        owner: root
        group: root
```

2. Edit the inventory file and add a workstation group. Add any Ubuntu remote server. Make sure to remember the IP address

```
jakeworkstation@WS: ~/JakeGestiada/HOA7
GNU nano 7.2                                  inventory.yaml
        ansible_host: 192.168.56.103
        ansible_user: jakeserver2

      centos1:
        ansible_host: 192.168.56.104
        ansible_user: jakecentos


  db_servers:
    hosts:
      ubuntu2:
        ansible_host: 192.168.56.103
        ansible_user: jakeserver2

      centos1:
        ansible_host: 192.168.56.104
        ansible_user: jakecentos


  file_servers:
    hosts:
      ubuntu1:
        ansible_host: 192.168.56.105
        ansible_user: jakeserver
```

3. Run the playbook. Describe the output.

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ ansible-playbook -i inventory.yaml site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]
ok: [centos1]
ok: [ubuntu1]

TASK [Update repository index (CentOS)] ****************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]
```

```
PLAY [workstations] ************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]

TASK [Install unzip] ***********************************************************************************
ok: [ubuntu2]

TASK [Install Terraform] *******************************************************************************
changed: [ubuntu2]

PLAY [web_servers] *************************************************************************************

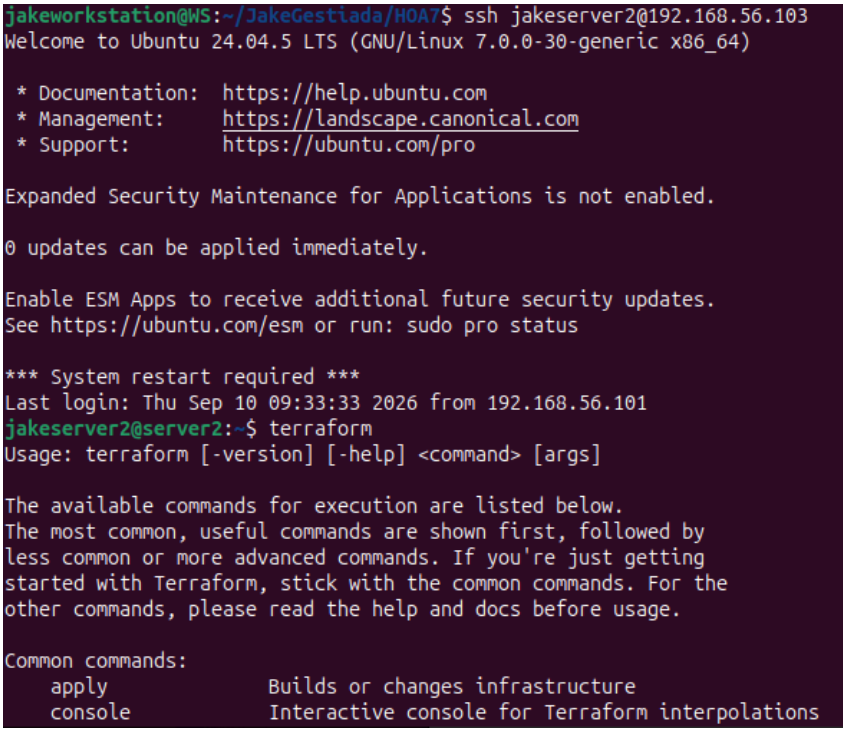
TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]
```

```
PLAY RECAP *********************************************************************************************
centos1                    : ok=8    changed=1    unreachable=0    failed=0    skipped=3    rescued=0
  ignored=0
ubuntu1                    : ok=7    changed=0    unreachable=0    failed=0    skipped=2    rescued=0
  ignored=0
ubuntu2                    : ok=11   changed=2    unreachable=0    failed=0    skipped=3    rescued=0
  ignored=0
```

Observation:

- The playbook installed unzip and downloaded Terraform to the remote Ubuntu server. The playbook finished successfully without errors.

4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output.



Observation:

- When I typed terraform on the Ubuntu server, the command worked and showed the Terraform output. This means Terraform was installed successfully.

Task 3: Create roles

1. Edit the site.yml. Configure roles as follows: (make sure to create a copy of the old site.yml file because you will be copying the specific plays for all groups

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ ls
files  inventory.yaml  site_old.yml  site.yml
jakeworkstation@WS:~/JakeGestiada/HOA7$ cat site.yml
---
- hosts: all
  become: true

  pre_tasks:

    - name: Update repository index (CentOS)
      ansible.builtin.dnf:
        update_cache: yes
      changed_when: false
      when: ansible_distribution == "CentOS"

    - name: Update package cache (Ubuntu)
      ansible.builtin.apt:
```

2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers, db_servers and workstations. For each directory, create directory and name it tasks

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ mkdir -p roles/base/tasks
jakeworkstation@WS:~/JakeGestiada/HOA7$ mkdir -p roles/web_servers/tasks
jakeworkstation@WS:~/JakeGestiada/HOA7$ mkdir -p roles/file_servers/tasks
jakeworkstation@WS:~/JakeGestiada/HOA7$ mkdir -p roles/db_servers/tasks
jakeworkstation@WS:~/JakeGestiada/HOA7$ mkdir -p roles/workstations/tasks
jakeworkstation@WS:~/JakeGestiada/HOA7$ find roles -type d
roles
roles/db_servers
roles/db_servers/tasks
roles/workstations
roles/workstations/tasks
roles/file_servers
roles/file_servers/tasks
roles/web_servers
roles/web_servers/tasks
roles/base
roles/base/tasks
```

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ tree roles
roles
├── base
│   └── tasks
│       └── main.yml
├── db_servers
│   └── tasks
│       └── main.yml
├── file_servers
│   └── tasks
│       └── main.yml
├── web_servers
│   └── tasks
│       └── main.yml
└── workstations
    └── tasks
        └── main.yml
```

Base:

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ cat roles/base/tasks/main.yml
---
- name: Update repository index (CentOS)
  ansible.builtin.dnf:
    update_cache: yes
  changed_when: false
  when: ansible_distribution == "CentOS"

- name: Update package cache (Ubuntu)
  ansible.builtin.apt:
    update_cache: yes
  changed_when: false
  when: ansible_distribution == "Ubuntu"
```

Workstation:

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ cat roles/workstations/tasks/main.yml
---
- name: Install unzip
  ansible.builtin.package:
    name: unzip
    state: present

- name: Install Terraform
  ansible.builtin.unarchive:
    src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
    dest: /usr/local/bin
    remote_src: yes
    mode: '0755'
    owner: root
    group: root
```

Web Servers:

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ cat roles/web_servers/tasks/main.yml
---
- name: Install Apache and PHP for Ubuntu servers
  ansible.builtin.apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: Install Apache and PHP for CentOS servers
  ansible.builtin.dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: Copy default HTML file for site
  tags:
    - apache
    - apache2
    - httpd
  ansible.builtin.copy:
    src: files/default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: '0644'
```

Database servers:

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ cat roles/db_servers/tasks/main.yml
---
- name: Install MariaDB package (CentOS)
  ansible.builtin.dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: MariaDB - Restarting/Enabling
  ansible.builtin.service:
    name: mariadb
    state: restarted
    enabled: true

- name: Install MariaDB package (Ubuntu)
  ansible.builtin.apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"
```

File servers:

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ cat roles/file_servers/tasks/main.yml
---
- name: Install Samba package
  ansible.builtin.package:
    name: samba
    state: latest
```

3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks.

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ touch roles/base/tasks/main.yml
jakeworkstation@WS:~/JakeGestiada/HOA7$ touch roles/web_servers/tasks/main.yml
jakeworkstation@WS:~/JakeGestiada/HOA7$ touch roles/file_servers/tasks/main.yml
jakeworkstation@WS:~/JakeGestiada/HOA7$ touch roles/db_servers/tasks/main.yml
jakeworkstation@WS:~/JakeGestiada/HOA7$ touch roles/workstations/tasks/main.yml
jakeworkstation@WS:~/JakeGestiada/HOA7$ find roles -type f
roles/db_servers/tasks/main.yml
roles/workstations/tasks/main.yml
roles/file_servers/tasks/main.yml
roles/web_servers/tasks/main.yml
roles/base/tasks/main.yml
```

4. Run the site.yml playbook and describe the output.

```
jakeworkstation@WS:~/JakeGestiada/HOA7$ ansible-playbook -i inventory.yaml site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu1]
ok: [ubuntu2]
ok: [centos1]

TASK [Update repository index (CentOS)] ****************************************************************
skipping: [ubuntu1]
skipping: [ubuntu2]
ok: [centos1]

TASK [Update package cache (Ubuntu)] *******************************************************************
skipping: [centos1]
ok: [ubuntu1]
ok: [ubuntu2]

PLAY [workstations] ************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]

TASK [Install unzip] ***********************************************************************************
ok: [ubuntu2]

TASK [Install Terraform] *******************************************************************************
ok: [ubuntu2]

PLAY [web_servers] *************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
```

```
PLAY [db_servers] ***************************************************************************************

TASK [Gathering Facts] **********************************************************************************
ok: [ubuntu2]
ok: [centos1]

TASK [Install MariaDB package (CentOS)] *****************************************************************
skipping: [ubuntu2]
ok: [centos1]

TASK [MariaDB - Restarting/Enabling] ********************************************************************
changed: [ubuntu2]
changed: [centos1]

TASK [Install MariaDB package (Ubuntu)] *****************************************************************
skipping: [centos1]
ok: [ubuntu2]

PLAY [file_servers] *************************************************************************************

TASK [Gathering Facts] **********************************************************************************
ok: [ubuntu1]

TASK [Install Samba package] ****************************************************************************
ok: [ubuntu1]

PLAY RECAP **********************************************************************************************
centos1                    : ok=8    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

ubuntu1                    : ok=7    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

ubuntu2                    : ok=11   changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

Observation:

- The site.yml playbook was run and roles were successfully applied to their appropriate servers. The PLAY RECAP showed that all the tasks had been successfully completed.

Github Link HOA7:

https://github.com/DamienJakeGestiada/CPE212_JakeGestiada/tree/main/HOA7

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Answer the following:

1. What is the importance of creating roles?
   - In Ansible, the concept of roles helps in keeping the project organized and makes it simpler to manage. It also allows me to reuse the same set of tasks for different servers without consolidating all the work into one playbook.

2. What is the importance of managing files?
   - Managing files is a very important thing that should be done, because one can create, copy, move, and arrange these files remotely on a server. With Ansible one can accomplish these things on several servers without the necessity of physically connecting to each server.

Conclusions/Learnings

- In this activity, I picked up a couple of things. I first found out that files in addition to whole roles could be handled with Ansible. I also got a hang of copying files to remote servers, unzipping them, installing and organizing tasks into different roles. All of that made a significant contribution towards knowing how Ansible could really transform server management into the kind of activity that is both easy and organized.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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